

Homework 8

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*All of my homework is available at <https://github.com/lacampanella233/Homework.git>.

Problem 1 (5-1). Let f be defined for all real numbers. Assume that for every pair of real numbers x and y ,

$$|f(x) - f(y)| \leq (x - y)^2.$$

Prove that f is a constant function.

Proof. If there exists $x < y$ such that $f(x) \neq f(y)$, then take a positive integer N such that $(y - x)^2 < N |f(y) - f(x)|$. Divide the interval $[x, y]$ into N equal subintervals, and let $x_0 = x, x_1, \dots, x_N = y$ be the endpoints of these subintervals, such that $x_k - x_{k-1} = (y - x)/N := \Delta$. Then for any $1 \leq k \leq N$,

$$|f(x_k) - f(x_{k-1})| \leq (x_k - x_{k-1})^2 = \Delta^2.$$

Summing these inequalities from $k = 1$ to $k = N$, we get

$$|f(y) - f(x)| \leq \sum_{k=1}^N |f(x_k) - f(x_{k-1})| \leq N \Delta^2 = \frac{(y - x)^2}{N}.$$

This contradicts our assumption that $(y - x)^2 < N |f(y) - f(x)|$. Therefore, $f(x) = f(y)$ for all $x, y \in \mathbb{R}$, and f is a constant function. $\square$

Problem 2 (5-4). Let $C_0, \dots, C_n$ be real constants, and suppose

$$C_0 + \frac{C_1}{2} + \dots + \frac{C_{n-1}}{n} + \frac{C_n}{n+1} = 0.$$

Prove that the equation

$$C_0 + C_1 x + \dots + C_{n-1} x^{n-1} + C_n x^n = 0$$

has at least one real root in $(0, 1)$.

Proof. Define

$$f(x) = C_0 x + \frac{C_1}{2} x^2 + \dots + \frac{C_{n-1}}{n} x^n + \frac{C_n}{n+1} x^{n+1}.$$

Then $f(0) = 0$ and $f(1) = C_0 + \frac{C_1}{2} + \dots + \frac{C_{n-1}}{n} + \frac{C_n}{n+1} = 0$. By Rolle's theorem, there exists some $c \in (0, 1)$ such that $f'(c) = 0$. Note that

$$f'(x) = C_0 + C_1 x + \dots + C_{n-1} x^{n-1} + C_n x^n.$$

Therefore, $f'(c) = 0$ implies that the equation $C_0 + C_1 x + \dots + C_{n-1} x^{n-1} + C_n x^n = 0$ has at least one real root in $(0, 1)$. $\square$

Problem 3 (5-5). Let f be defined and differentiable for all $x > 0$, and assume that $f'(x) \rightarrow 0$ as $x \rightarrow +\infty$. Define

$$g(x) = f(x+1) - f(x).$$

Prove that $g(x) \rightarrow 0$ as $x \rightarrow +\infty$.

Proof. By the mean value theorem, for each $x > 0$, there exists some $c_x \in (x, x+1)$ such that

$$g(x) = f(x+1) - f(x) = f'(c_x)(x+1-x) = f'(c_x).$$

Since $c_x \rightarrow +\infty$ as $x \rightarrow +\infty$ and $f'(x) \rightarrow 0$ as $x \rightarrow +\infty$, we have $g(x) = f'(c_x) \rightarrow 0$ as $x \rightarrow +\infty$. $\square$

Problem 4 (5-6). Assume:

- (a) f is continuous on $x \geq 0$;
- (b) $f'(x)$ exists for $x > 0$;
- (c) $f(0) = 0$;
- (d) f' is increasing.

Let

$$g(x) = \frac{f(x)}{x} \quad (x > 0).$$

Prove that g is increasing.

Proof. Suppose the contrary, that is, there exist $0 < a < b$ such that $g(a) > g(b)$. By the mean value theorem, there exists some $c \in (a, b)$ such that

$$f(b) - f(a) = f'(c)(b-a).$$

Since $g(a) > g(b)$, we have

$$f'(c) = \frac{f(b) - f(a)}{b-a} = \frac{g(b)b - g(a)a}{b-a} < \frac{g(a)b - g(a)a}{b-a} = g(a).$$

By the mean value theorem again, there exists some $d \in (0, a)$ such that

$$f(a) - f(0) = f'(d)(a-0) \implies f'(d) = g(a).$$

Since f' is increasing and $d < c$, we have $f'(d) \leq f'(c)$, which contradicts $f'(d) = g(a) > f'(c)$. Therefore, g is increasing. $\square$

Problem 5 (5-9). Let f be a continuous real-valued function on $\mathbb{R}$. Assume that $f'(x)$ exists for every $x \neq 0$, and that $f'(x) \rightarrow 3$ as $x \rightarrow 0$. Question: does $f'(0)$ necessarily exist?

Solution. Yes. We will prove that

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = 3.$$

First of all, by $\lim_{x \rightarrow 0} f'(x) = 3$, we know that f is differentiable at some neighbourhood of 0. For any $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $0 < |x| < \delta$, we have $|f'(x) - 3| < \varepsilon$. Then for all $0 < |x| < \delta$, by the mean value theorem, there exists some $c_x \in (0, x)$ or $(x, 0)$ such that

$$\frac{f(x) - f(0)}{x} = f'(c_x) \in (3 - \varepsilon, 3 + \varepsilon).$$

This implies that $\lim_{x \rightarrow 0} (f(x) - f(0))/x = 3$, and thus $f'(0)$ exists and equals 3.

Problem 6 (5-11). Let f be defined in a neighborhood of x , and assume $f''(x)$ exists. Prove that

$$\lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} = f''(x).$$

Give an example showing that this limit may still exist even when $f''(x)$ does not exist.

Proof. Since $f''(x)$ exists, by the Taylor's formula with the Peano's form of the remainder, we have

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)}{2}h^2 + o(h^2).$$

Then

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} \\ &= \lim_{h \rightarrow 0} \frac{1}{h^2} \left[\left(f(x) + f'(x)h + \frac{f''(x)}{2}h^2 + o(h^2) \right) + \left(f(x) - f'(x)h + \frac{f''(x)}{2}h^2 + o(h^2) \right) - 2f(x) \right] \\ &= \lim_{h \rightarrow 0} \frac{f''(x)h^2 + o(h^2)}{h^2} \\ &= f''(x). \end{aligned}$$

□

Problem 7 (5-17). Let f be a real-valued function on $[-1, 1]$ that is three times differentiable, and suppose

$$f(-1) = 0, \quad f(0) = 0, \quad f(1) = 1, \quad f'(0) = 0.$$

Prove that there exists some $x \in (-1, 1)$ such that

$$f^{(3)}(x) \geq 3.$$

Note: for the function $\frac{1}{2}(x^3 + x^2)$, equality is attained.

Proof. Denote $g(x) = f(x) - \frac{1}{2}(x^3 + x^2)$, then $g(-1) = g(0) = g(1) = g'(0) = 0$, and $g^{(3)}(x) = f^{(3)}(x) - 3$.

Suppose the contrary, that is, $g^{(3)}(x) < 0$ for all $x \in (-1, 1)$, then g'' is strictly decreasing on $(-1, 1)$. Then, either $g''(x) > 0$ for all $x \in (0, 1)$ or $g''(x) < 0$ for all $x \in (-1, 0)$.

If $g''(x) > 0$ for all $x \in (0, 1)$, then g' is strictly increasing on $(0, 1)$. Since $g'(0) = 0$, we have $g'(x) > 0$ for all $x \in (0, 1)$. This contradicts the fact that $g(1) = 0$.

If $g''(x) < 0$ for all $x \in (-1, 0)$, then we can get a contradiction in a similar way. Therefore, there exists some $x \in (-1, 1)$ such that $g^{(3)}(x) \geq 0$, which implies that $f^{(3)}(x) \geq 3$. □

Problem 8 (5-22). Let f be a real function on $(-\infty, \infty)$. If $f(x) = x$, then x is called a fixed point of f .

(a) If f is differentiable and $f'(t) \neq 1$ for every real t , prove that f has at most one fixed point.

(b) Prove that

$$f(t) = t + (1 + e^t)^{-1}$$

has no fixed point, although for every real t one has $0 < f'(t) < 1$.

(c) However, if there is a constant $A < 1$ such that for every real t ,

$$|f'(t)| \leq A,$$

prove that f has a fixed point x , and moreover

$$x = \lim_{n \rightarrow \infty} x_n,$$

where x_1 is any real number and for $n = 1, 2, 3, \dots$,

$$x_{n+1} = f(x_n).$$

(d) Prove that the method in (c) can be implemented along the zig-zag path

$$(x_1, x_2) \rightarrow (x_2, x_2) \rightarrow (x_2, x_3) \rightarrow (x_3, x_3) \rightarrow (x_3, x_4) \rightarrow \dots$$

Proof. (a) Suppose the contrary, if $f(x) = x$ and $f(y) = y$ for some $x \neq y$, then by the mean value theorem, there exists some $c \in (x, y)$ such that

$$f'(c) = \frac{f(y) - f(x)}{y - x} = 1,$$

which contradicts the assumption that $f'(t) \neq 1$ for every real t . Therefore, f has at most one fixed point.

(b) Since $(1 + e^t)^{-1} \neq 0$ for all real t , we have $f(t) = t + (1 + e^t)^{-1} \neq t$ for all real t . Therefore, f has no fixed point.

Note that

$$f'(t) = 1 - \frac{e^t}{(1 + e^t)^2} > 1 - 1 = 0,$$

and obviously $f'(t) < 1$ for all real t . Therefore, $0 < f'(t) < 1$ for all real t .

(c) Suppose the contrary, that is $f(x) \neq x$ for all $x \in \mathbb{R}$. Then by the continuity of f , we have either $f(x) > x$ for all $x \in \mathbb{R}$ or $f(x) < x$ for all $x \in \mathbb{R}$. We only consider the former case, and the latter case can be treated similarly.

If $f(x) > x$ for all $x \in \mathbb{R}$, then for any $x > 0$, there exists some $\xi_x \in (0, x)$, such that

$$f(x) - f(0) = f'(\xi_x)(x - 0) \leq Ax.$$

So we conclude that $x < f(x) \leq Ax + f(0)$ for all $x > 0$, which contradicts the fact that $A < 1$. Therefore, f has a fixed point x .

Next we will prove that $x = \lim_{n \rightarrow \infty} x_n$. For any $n \geq 1$, we have (by the mean value theorem)

$$|x_{n+2} - x_{n+1}| = |f(x_{n+1}) - f(x_n)| = |f'(\xi_n)(x_{n+1} - x_n)| \leq A|x_{n+1} - x_n|.$$

This implies (maybe with a little effort) that $\{x_n\}$ is a Cauchy sequence, and hence converges to some limit x . Then, take $n \rightarrow \infty$ in the equation $x_{n+1} = f(x_n)$, we get $x = f(x)$, which implies that x is the fixed point of f . Therefore, $x = \lim_{n \rightarrow \infty} x_n$.

(d) Actually, the method in (c) is exactly the same as the zig-zag path described in (d). Therefore, the conclusion in (c) also holds for the zig-zag path. $\square$